

Multi-Criteria Decision Analysis (MCDA) Framework

Applied Case Studies, Optimization Models, and Methodological
Benchmarks

Decision-Making Fundamentals

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Abstract

This document serves as the formal specification and case study benchmark suite for evaluating decision-making models under conditions of uncertainty, structural interdependence, and multi-attribute preference structures. Four comprehensive cases are explored: (i) production planning under severe demand volatility combined with objective and subjective weighting methods (Entropy and SWARA); (ii) optimization-driven derivation of priority vectors from Pairwise Comparison Matrices (LSM, DEAHP, and LPM); (iii) structural cause-effect and feedback network modeling using AHP, ANP, and DEMATEL; and (iv) an end-to-end comparative benchmark of outranking and distance-based multi-criteria ranking algorithms (TOPSIS, VIKOR, ELECTRE I, PROMETHEE I/II with GAIA visual modeling) coupled with rank aggregation techniques.

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1 Case Study I: Decision Making Under Uncertainty and Attribute Weighting

1.1 Problem Context: Manufacturing Product Selection

An industrial manufacturing firm faces strategic product line selection under uncertain market absorption capacity. Six potential product variants (P_1 through P_6) have been engineered, with net operational payoffs contingent upon three macroeconomic demand scenarios: Low Demand (S_1), Medium Demand (S_2), and High Demand (S_3).

Table 1: Payoff Matrix under Stochastic Market States (Values in Million Tomans)

Alternative Product	Low Demand (S_1)	Medium Demand (S_2)	High Demand (S_3)
<i>State Probability (P_j)</i>	<i>0.31</i>	<i>0.45</i>	<i>0.24</i>
Product 1 (P_1)	1,050	1,500	2,100
Product 2 (P_2)	1,350	750	2,400
Product 3 (P_3)	1,650	1,500	2,850
Product 4 (P_4)	1,200	750	1,800
Product 5 (P_5)	1,500	1,125	2,700
Product 6 (P_6)	1,800	1,500	2,850

Research Requirements:

- Decisions under Complete Uncertainty:** Determine the optimal product portfolio assuming state probabilities are entirely unknown, utilizing:
 - Maximax (Optimistic criterion)
 - Maximin / Wald (Pessimistic criterion)
 - Hurwicz Alpha Criterion ($\alpha \in [0, 1]$)
 - Savage Minimax Regret (Opportunity loss criterion)
 - Principle of Insufficient Reason (Laplace criterion)
- Decisions under Risk:** Apply the Bayes / Expected Monetary Value (EMV) decision rule based on the calibrated state probabilities ($P_1 = 0.31, P_2 = 0.45, P_3 = 0.24$).
- Sensitivity and Discrepancy Analysis:** The manufacturing director asserts that marketing estimates systematically underestimate revenues for Products 1–3 by 30% and overestimate Products 4–6 by 30%. Recalculate all models under this revised scenario and document decision shifts.

1.2 Decision Matrix Normalization and Weighting

A corporate facility location problem considers six alternative municipal zones evaluated over five criteria: Purchase Price (C_1 , Cost), Market Access (C_2 , Cost/Distance), Traffic Congestion (C_3 , Cost), Green Space / Parks (C_4 , Benefit), and Public Transit Access (C_5 , Benefit).

Methodological Implementation:

- Convert qualitative linguistics into quantitative scales using a standard 10-point scale.

Table 2: Municipal Zone Evaluation Matrix

Zone	C_1 : Price	C_2 : Market Access	C_3 : Traffic	C_4 : Parks	C_5 : Transit
Zone 1	3,000	Close	High	5	High
Zone 2	6,000	Far	Moderately High	3	Very High
Zone 3	6,750	Moderately Close	Medium	4	Moderately Low
Zone 4	2,250	Very Close	Low	2	Medium
Zone 5	5,250	Moderately Far	High	6	Low
Zone 6	4,500	Very Close	Low	4	Moderately High
Target Profile	3,750	Close	Low	4	Medium

2. Execute normalization using: (i) Euclidean (Vector) normalization, (ii) Linear sum-based normalization, (iii) Linear max-min normalization, (iv) Fuzzy/Interval range normalization, (v) Standard Gaussian (z-score) normalization, and (vi) Distance-to-target (Centroid) normalization.
3. Compute criterion weights using:
 - **Shannon Entropy:** Unmodified objective weights and modified subjective-objective weights with prior beliefs $\mathbf{w}_0 = [0.15, 0.05, 0.20, 0.10, 0.50]^T$.
 - **SWARA Method:** Based on a public preference poll of 225 citizens allocating votes across criteria: $C_1 = 180$, $C_2 = 12$, $C_3 = 18$, $C_4 = 6$, and $C_5 = 9$.

2 Case Study II: Optimization-Based Priority Vector Derivation from Pairwise Comparison Matrices (PCM)

2.1 Pairwise Comparison Matrix Specification

Consider a multi-criteria prioritization problem involving four fundamental decision criteria with the following calibrated reciprocal matrix $\mathbf{A}$:

$$\mathbf{A} = \begin{bmatrix} 1.000 & 1.000 & 1.500 & 1.000 \\ 1.000 & 1.000 & 1.500 & 0.500 \\ 0.667 & 0.667 & 1.000 & 1.000 \\ 1.000 & 2.000 & 1.000 & 1.000 \end{bmatrix} \quad (1)$$

2.2 Analytical and Mathematical Programming Formulations

1. Spectral and Heuristic Approaches:

- Compute the principal priority vector $\mathbf{w}$ using Saaty's Eigenvector Method ($\mathbf{A}\mathbf{w} = \lambda_{\max}\mathbf{w}$). Compute the Consistency Index (CI) and Consistency Ratio (CR).
- Approximate the priority vector via row-sum, column-normalization, and geometric mean approximations, benchmarking deviations against the principal eigenvector.

2. Least Squares Method (LSM): Formulate and solve the non-linear optimization model minimizing residual squared errors:

$$\min \sum_{i=1}^n \sum_{j=1}^n \left(a_{ij} - \frac{w_i}{w_j} \right)^2 \quad (2)$$

$$\text{s.t.} \quad \sum_{i=1}^n w_i = 1, \quad w_i > 0 \quad \forall i = 1, \dots, n \quad (3)$$

3. Data Envelopment Analysis Analytic Hierarchy Process (DEAHP): Derive local weights by treating each criterion row as a Decision Making Unit (DMU) to maximize the efficiency ratio subject to bounded virtual weights.

4. Linear Programming Method (LPM): Solve a two-phase linear goal program minimizing absolute deviations:

$$\min \sum_{i=1}^n \sum_{j=1}^n \left(d_{ij}^+ + d_{ij}^- \right) \quad (4)$$

$$\text{s.t.} \quad w_i - a_{ij}w_j - d_{ij}^+ + d_{ij}^- = 0 \quad \forall i \neq j \quad (5)$$

$$\sum_{i=1}^n w_i = 1, \quad w_i \geq \epsilon, \quad d_{ij}^+, d_{ij}^- \geq 0 \quad (6)$$

3 Case Study III: Hierarchical, Network, and Causal Structural Decision Frameworks

3.1 System Architecture: Strategic Vehicle Procurement

The problem addresses the multi-attribute evaluation and selection among three competing commercial vehicle platforms (V_1, V_2, V_3) evaluated against four primary criteria and four subordinate technical sub-criteria:

- **Price (C_1):** Total acquisition and capital expenditure [Cost].
- **Fuel Consumption (C_2):** Operational fuel consumption per 100 km [Cost].
- **Safety (C_3):** [Benefit], partitioned into:
 - C_{31} : Structural Body Safety
 - C_{32} : Airbag and Restraint System Capacity
- **Technical Performance (C_4):** [Benefit], partitioned into:
 - C_{41} : Engine Displacement and Power Capacity
 - C_{42} : Maximum Sustainable Velocity

3.2 Multi-Model Implementation Pipeline

1. Analytic Hierarchy Process (AHP): Construct the complete three-tier hierarchy (Goal $\rightarrow$ Criteria/Sub-criteria $\rightarrow$ Alternatives). Populate consistent pairwise comparison matrices at all decision levels, verify consistency ($CR < 0.1$), and synthesize the global priority vector using *Expert Choice*.

2. Analytic Network Process (ANP): Extend the hierarchical model into a network system accounting for feedback loops between alternatives and criteria (e.g., how the presence of specific alternatives alters the subjective importance of safety versus cost). Synthesize the unweighted supermatrix, weighted supermatrix, and limiting supermatrix using *Super Decisions*.

3. DEMATEL (Decision Making Trial and Evaluation Laboratory): Analyze mutual interdependence and causal impact across the six criteria ($C_1, C_2, C_{31}, C_{32}, C_{41}, C_{42}$).

- Construct the direct-relation matrix $\mathbf{Z}$ based on aggregated expert consensus.
- Normalize the direct influence matrix: $\mathbf{X} = \frac{\mathbf{Z}}{\max_{1 \leq i \leq n} \sum_{j=1}^n z_{ij}}$.
- Derive the total relation matrix $\mathbf{T} = \mathbf{X}(\mathbf{I} - \mathbf{X})^{-1}$.
- Compute causal vectors ($R + D$ for prominence and $R - D$ for net impact) and plot the Cause-Effect Causal Diagram.

4 Case Study IV: Multi-Criteria Ranking Benchmark and Rank Aggregation

4.1 Experimental Benchmark Setup

Applying the performance vectors and criterion weights obtained from Case Study III, the three vehicle candidates (V_1, V_2, V_3) undergo rigorous ranking under four distinct MCDM computational paradigms.

4.2 Algorithmic Implementations

1. Distance-Based Paradigm (TOPSIS): Evaluate relative closeness to the Positive Ideal Solution (PIS, A^+) and Negative Ideal Solution (NIS, A^-) using Euclidean distance metrics:

$$CC_i = \frac{d_i^-}{d_i^+ + d_i^-}, \quad CC_i \in [0, 1] \quad (7)$$

2. Compromise Programming Paradigm (VIKOR): Compute maximum group utility (S_i) and individual minimum regret (R_i):

$$Q_i = v \frac{S_i - S^*}{S^- - S^*} + (1 - v) \frac{R_i - R^*}{R^- - R^*} \quad (8)$$

Verify both Acceptable Advantage ($Q(A^{(2)}) - Q(A^{(1)}) \geq DQ$) and Acceptable Stability in Decision Making.

3. Outranking Dominance Paradigm (ELECTRE I): Construct the Concordance Index (c_{ik}) and Discordance Index (d_{ik}) matrices. Apply threshold filters ($\bar{c}, \bar{d}$) to establish the kernel of non-dominated alternatives.

4. Preference Flow Paradigm (PROMETHEE I & II and GAIA Analysis): Select generalized preference functions (Type V: Linear with indifference, or Type III: V-shape) for criteria.

- Compute positive leaving flows (ϕ^+) and negative entering flows (ϕ^-) for partial preorder (PROMETHEE I).
- Derive net outranking flows ($\phi = \phi^+ - \phi^-$) for complete ordering (PROMETHEE II).
- Execute Visual PROMETHEE GAIA (Geometrical Analysis for Interactive Aid) principal component decomposition to visualize criteria conflicts and alternative positioning on the (π, u, v) plane.

4.3 Consensus and Rank Aggregation Benchmark

Collate the independent rank orders generated across all methodologies: AHP, ANP, DEMATEL-weighted, TOPSIS, VIKOR, ELECTRE I, and PROMETHEE II. Synthesize a unified meta-ranking using:

1. Mean Rank Method
2. Borda Count Algorithm
3. Copeland Pairwise Tournament Method
4. Spearman Rank Correlation Matrix to quantify methodological consensus.